Syntactic Priming Facilitates L2 Learning of the Mandarin BEI-passive: Comparing the Role of Guessing and Repetition in Oral Production

Hailong Chen & Katherine Messenger (Lancaster University)
h.chen52@lancaster.ac.uk

Background. *Syntactic priming*, the tendency for speakers to mirror the syntactic patterns of recently encountered sentences when producing new utterances (Bock, 1986), has been shown to facilitate L2 acquisition and the development of grammatical knowledge (Jackson, 2018). Language learning via priming is commonly attributed to implicit, error-based learning mechanisms (Chang et al., 2006), whereby prediction errors (i.e., the mismatch between expected and actual input) drive adaptation. A growing body of research suggests that forced prediction (e.g., through a "guessing game" task) could lead to stronger syntactic adaptation than simple repetition (e.g., Grüter et al., 2021; Kim et al., 2025). However, most existing evidence comes from written production tasks and limited syntactic structures (e.g., datives). Thus, it remains unclear whether the benefits of prediction-based priming extend to spoken production and to typologically distinct structures such as the Mandarin BEI-passive. This construction is a key yet challenging grammatical construction for L2 learners, who often avoid using it (Yang, 2024), highlighting the need for effective pedagogical approaches to support its acquisition.

Method. The study adopted a pretest-posttest design to examine whether forced prediction enhances oral production of the Mandarin BEI-passive. Between pre-/post-tests, adult L2 Chinese learners and L1 Chinese speakers completed an oral picture-description priming task, consisting of prime trials paired with corresponding target trials, in either a Guessing or Repetition condition (Fig.1). In Guessing (L2: n=48; L1: n=50), participants first predicted how a virtual partner (i.e., a native Mandarin speaker, "Li Ming") would describe a picture. They then heard the actual prime sentence and judged whether it matched their prediction. In Repetition (L2: n=46; L1: n=50), participants simply heard and repeated the prime sentence. In all conditions, each prime was followed by a target trial, where participants described a picture aloud. In both pre-/post-tests, participants also completed a picture-description task, producing pre-/post-exposure target descriptions without any priming intervention.

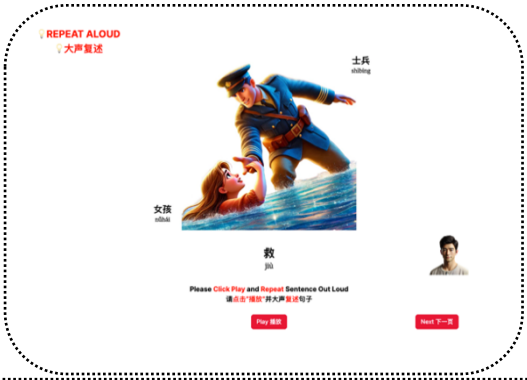
Results. The results showed participants produced significantly more BEI-passives after BEI-passive primes than after Active primes ($p < .001$; Fig.2a), suggesting a robust syntactic priming effect during the immediate priming phase. L2 learners produced more BEI-passives overall than L1 speakers ($p < .001$). A significant Prime × Condition interaction ($p < .001$) showed that immediate priming was stronger in the Repetition condition for both groups. Participants produced more BEI-passives at posttest than pretest ($p < .001$; Fig.2b), showing syntactic adaptation. However, Condition did not modulate these cumulative gains (all *ps* > .63): both Guessing and Repetition groups demonstrated comparable increases over time. Group also showed a significant main effect ($p < .001$), with L2 learners producing more BEI-passives overall than L1 speakers. Importantly, the Phase × Group interaction was significant ($p = .002$), reflecting a tendency for L2 learners to show slightly larger learning gains than L1 speakers. Within-group analyses confirmed that both L2 and L1 speakers showed robust syntactic adaptation (*ps* < .01).

Discussion. The findings provide evidence that syntactic priming facilitates L2 learning of the Mandarin BEI-passive, with clear increases both in immediate use and from pretest to posttest. L2 learners showed long-term adaptation, consistent with accounts proposing their sensitivity to error-based learning in L2 processing. However, forced prediction through Guessing did not enhance cumulative learning, and even reduced immediate priming compared to Repetition in the oral modality. Thus, we find no evidence that guessing facilitates L2 syntactic learning in oral production. Still, the substantial cumulative gains observed for the L2 group in both Guessing and Repetition conditions highlight the potential of syntactic priming as a practical tool for promoting L2 structure learning.

References: Bock, J. K. (1986). Syntactic persistence in language production. *Cognitive Psychology, 18*(3), 355-387. Chang, F., Dell, G. S., & Bock, K. (2006). Becoming syntactic. *Psychological Review, 113*(2), 234-272. Grüter, T., Zhu, Y. A., & Jackson, C. N. (2021). *Forcing prediction increases priming and adaptation in second language production.* In E. Kaan & T. Grüter (Eds.), Bilingual Processing and Acquisition (Vol. 12, pp. 207-232). John Benjamins Publishing Company. Jackson, C. N. (2018). Second language structural priming: A critical review and directions for future research. *Second Language Research, 34*(4), 539-552. Kim, Y., Lee, H., Han, S., Yu, M., & Kim, K. (2025). *Prediction errors facilitate L2 learning: Evidence from structural priming and prediction error responses.* Poster presented at the 50th Annual Boston University Conference on Language Development (BUCLD 50), November 6-9.

Figure 1. Prime trial for Repetition and Guessing condition.

Repetition:



Guessing:

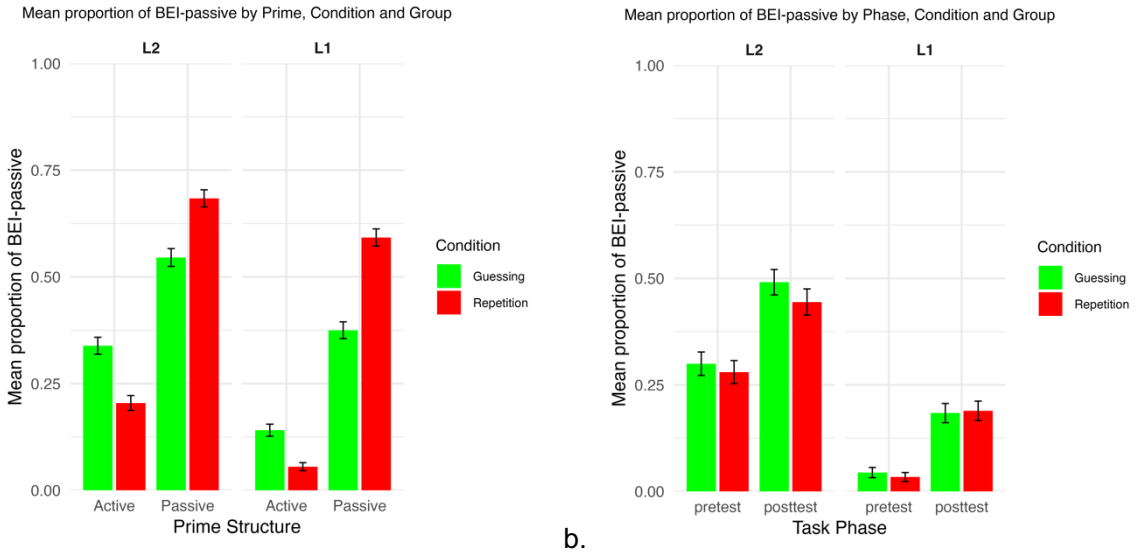



Figure 2. Mean proportion of target responses for immediate priming phase (a) and pre-/post-test (b).